STM32F103VET6 + BMI270 (SPI1) + SpeedyBee ELRS (USART1)

MODULE INTERCONNECT — header IDs are NOT MCU package pin numbers

A1
STM32F103VET6_DevKit

S1 S2 S3 S4 BOOT0 BOOT1

GND SWDIO SWCLK 3V3 BOOT0 BOOT1

USB UP

LO01 GND
LO01 GND
LO02 VBAT
LI02 +3V3
LO03 +5V
LI03 5V
LO04 E0
LI04 E1
LO05 E2
LI05 E3
LO06 E4
LI06 E5
LO07 E6
LI07 C13
LO08 C14
LI08 C15
LO09 C0
LI09 C1
LO10 C2
LI10 C3
LO11 IMU_INT1 A0
LI11 A2
LO12 A1
LI12 A3
LO13 IMU_CS_N A4
LI13 SPI1_SCK A5
LO14 SPI1_MISO A6
LI14 SPI1_MOSI A7
LO15 C4
LI15 C5
LO16 B0
LI16 B1
LO17 B2
LI17 E7
LO18 E8
LI18 E9

RI001 GND
RI01 GND
RO02 3V3
RI02 3V3
RO03 GND
RI03 GND
RO04 B8
RI04 B9
RO05 B6
RI05 B7
RO06 B4
RI06 B5
RO07 D7
RI07 B3
RO08 D5
RI08 D6
RO09 D3
RI09 D4
RO10 D1
RI10 D2
RO11 C12
RI11 D0
RO12 C11
RI12 C10
RO13 A15
RI13 A14
RO14 A13
RI14 A12
RO15 A11
RI15 A10 CRSF_RX_MCU
RO16 A9 CRSF_TX_MCU
RI16 A8
RO17 C9
RI17 C7
RO18 C8
RI18 C6

STM32F103VET6 DEV KIT

LEFT 2x18 / RIGHT 2x18

BOTTOM 2x10 / SWD + BOOT

O = OUTER / I = INNER

E10 E11 E12 E13 E14 E15 B10 B11 B12 B13 B14 B15 D8 D9 D10 D11 D12 D13 D14 D15

BI01 BO01 BI02 BO02 BI03 BO03 BI04 BO04 BI05 BO05 BI06 BO06 BI07 BO07 BI08 BO08 BI09 BO09 BI10 BO10

U1: 3.3V input compatibility / onboard regulator must be verified on actual V1.3 module.

U1
BMI270_DevKit_V1_3

BMI270
V1.3

8 SDX
9 SCX
10 INT2

1 VCC +3V3
2 GND GND
3 ADO/MISO SPI1_MISO
4 SDA/MOSI SPI1_MOSI
5 SCL/SCLK SPI1_SCK
6 CS IMU_CS_N
7 INT1 IMU_INT1

SDX / SCX unused: disable auxiliary interface. INT2 unused.

R1: CS pull-up; omit if already fitted on module.

+3V3
10k
R1
IMU_CS_N

+3V3
C1
100n
GND

+3V3
C2
1u
GND

U2
SB_Nano_2G4RX_Physical

RX TX 5V GND
1 2 3 4

CRSF_TX_MCU
CRSF_RX_MCU
+5V
GND

+5V
C3
10u
GND

C1/C2: near U1 VCC; C3: near U2 supply.

SPI1 default pins: PA4 GPIO CS; PA0 EXTI0. USART1 default PA9 TX / PA10 RX.

Firmware: SPI 4-wire mode 0 or 3; start at 1MHz. CRSF non-inverted full-duplex.

BOOT0 low for normal boot. Footprints intentionally unset: module dimensions unverified.

J1
Conn_01x02
+5V 1
GND 2

PWR_FLAG +5V
PWR_FLAG +3V3
PWR_FLAG GND

POWER: J1 regulated 5V input; A1 onboard regulator supplies 3V3 (verify rating).

Use one 5V source. Check USB backfeed path before connecting external 5V.

Flags assert external 5V, onboard 3V3 and common GND; they are not regulators.

Sheet: / File: kicad.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/1